

Logistic Regression

David Puelz

The University of Austin

The basic logistic regression model

Interpreting the model results

Evaluating the model

Logistic regression with multiple predictors

Interactions in logistic regression

Hypothesis testing

Applications



Have an account?

Sign in

DATING DESERVES BETTER

On OkCupid, you're more than just a photo. You have stories to tell, and passions to share, and things to talk about that are more interesting than the weather. Get noticed for who you are, not what you look like. Because you deserve what dating deserves: better.

JOIN **okc**

By clicking Join, you agree to our [Terms of Service](#). Learn about how we process and use your data in our [Privacy Policy](#) and how we use cookies and similar technology in our [Cookie Policy](#).

GET THE APP





The OkCupid data set

- The OkCupid data set contains information about 59946 profiles from users of the OkCupid online dating service.
- We have data on user age, height, sex, income, sexual orientation, education level, body type, ethnicity, and more.
- OkCupid often publishes their own analyses—see <https://theblog.okcupid.com/tagged/data>.
- Let's see if we can predict the sex/gender of the user based on their height.



What's wrong with this regression?

$$\widehat{\text{sex}} = \hat{\beta}_0 + \hat{\beta}_1 \cdot \text{height}$$



What's wrong with this regression?

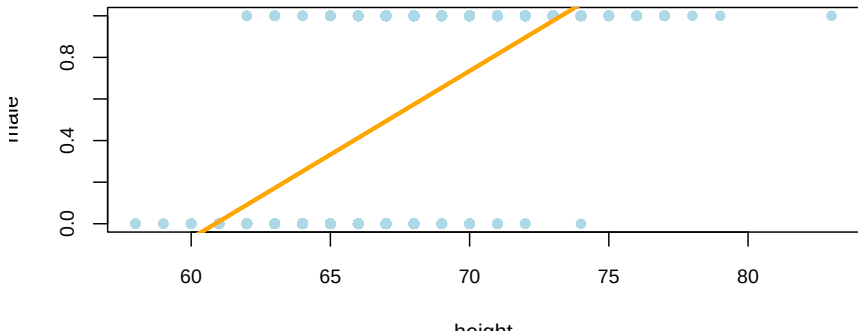
$$\widehat{\text{sex}} = \hat{\beta}_0 + \hat{\beta}_1 \cdot \text{height}$$

The Y variable here is **categorical** (two levels—male or female), so regular linear regression might not be the best choice.



But what if we just do it anyway?

We created a dummy: male = 1, female = 0. Fitting a line:



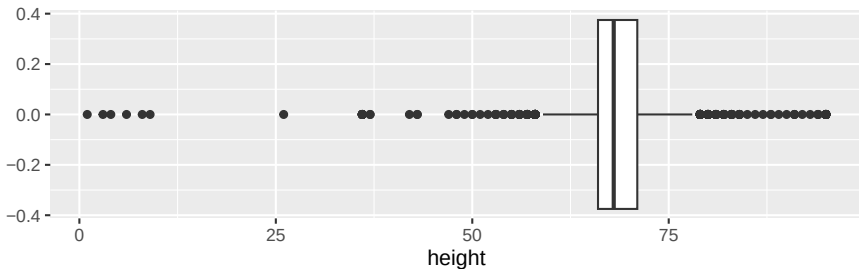
A line is a spectacularly bad fit. And what does it mean to predict male = 0.7 (or 1.2)?



Cleaning the data

There are some weird values for height:

```
ggplot(profiles, aes(x = height)) + geom_boxplot()
```



Cleaning the data



Consider only heights between 55 and 80 inches (4'7" and 6'8"). This excludes only 117 cases out of 59946.

```
my.profiles <- profiles %>% filter(height >= 55 & height <= 80)
```



The idea behind logistic regression

- Instead of predicting male/female, predict the *probability* that they are male.
- One level is “success” ($Y = 1$), the other “failure” ($Y = 0$). Here male = success.
- Fit a curve that is always between 0 and 1.



- “Even (1/1) odds” $\Rightarrow$ probability of success is $1/2$



Odds

- “Even (1/1) odds” $\Rightarrow$ probability of success is $1/2$
- “2/1 odds” $\Rightarrow$ probability is $2/3$



- “Even (1/1) odds” $\Rightarrow$ probability of success is $1/2$
- “2/1 odds” $\Rightarrow$ probability is $2/3$
- In general: odds of something with probability p are $p/(1 - p)$



Odds example

During March Madness, a \$100 bet on San Diego State paid \$360—odds to win of $1/3.6 = 5/18$. Implied *probability* that SDSU wins?



Odds example

During March Madness, a \$100 bet on San Diego State paid \$360—odds to win of $1/3.6 = 5/18$. Implied *probability* that SDSU wins?

$$\frac{p}{1-p} = \frac{1}{3.6} \Rightarrow p = \frac{1}{4.6} \approx 0.22$$



The logistic regression model

Logistic regression models the **log odds** of success as a linear function of X :

$$\log \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 X$$

This fits an S-shaped curve (bounded between 0 and 1).



Let's try it

```
model <- glm(male ~ height, data=my.profiles, family=binomial)
summary(model)
```



Let's try it

Call:

```
glm(formula = male ~ height, family = binomial, data = my.profiles)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-44.44861	0.35751	-124	<2e-16 ***
height	0.66190	0.00529	125	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 80654 on 59825 degrees of freedom
Residual deviance: 44637 on 59824 degrees of freedom
AIC: 44641

Number of Fisher Scoring iterations: 6



How to interpret the curve?

The regression says

$$\log(\text{odds}) = \log \left(\frac{P(\text{male})}{1 - P(\text{male})} \right) = -44.45 + 0.66 \cdot \text{height}.$$



How to interpret the curve?

The regression says

$$\log(\text{odds}) = \log\left(\frac{P(\text{male})}{1 - P(\text{male})}\right) = -44.45 + 0.66 \cdot \text{height}.$$

Solving for $P(\text{male})$:

$$\widehat{P(\text{male})} = \frac{e^{-44.45+0.66 \cdot \text{height}}}{1 + e^{-44.45+0.66 \cdot \text{height}}}$$



Making predictions

```
predict(model, data.frame(height=69), type="response")
```

1

0.77

So $\frac{e^{-44.45+0.66 \cdot 69}}{1+e^{-44.45+0.66 \cdot 69}} = 0.77$: we predict someone who is 5'9" has about 77% chance of being male.

The basic logistic regression model

Interpreting the model results

Evaluating the model

Logistic regression with multiple predictors

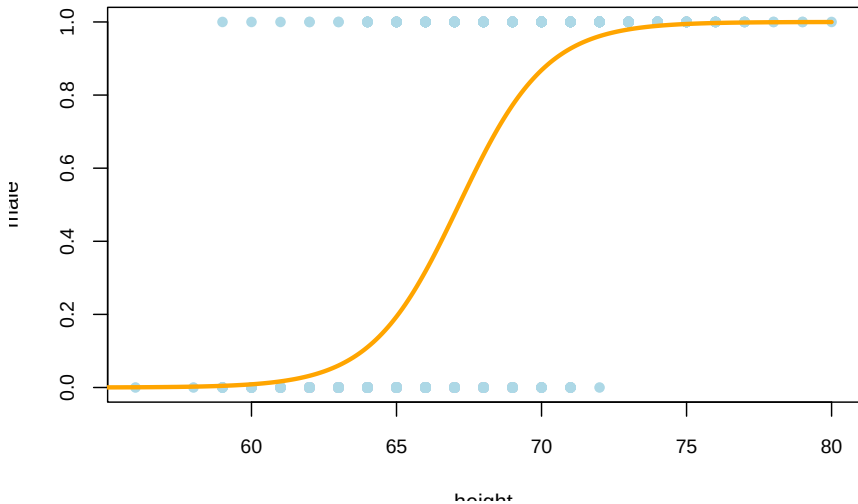
Interactions in logistic regression

Hypothesis testing

Applications



Visualizing the model





Interpreting the coefficients

$$\log \left(\frac{P(\text{male})}{1 - P(\text{male})} \right) = -44.45 + 0.66 \cdot \text{height}.$$

- When height = 0, predicted log odds = -44.45 ; predicted probability $\approx 0\%$.



Interpreting the coefficients

$$\log \left(\frac{P(\text{male})}{1 - P(\text{male})} \right) = -44.45 + 0.66 \cdot \text{height}.$$

- When height = 0, predicted log odds = -44.45 ; predicted probability $\approx 0\%$.
- When height increases by 1 inch, log odds increase by 0.66.



Interpreting the coefficients

Rewriting: Predicted odds of male = $e^{-44.45+0.66 \cdot \text{height}}$. Increasing height by 1 inch *multiplies* the odds by $e^{0.66} = 1.94$; i.e., increases odds by 94%.



Interpreting the coefficients

Rewriting: Predicted odds of male = $e^{-44.45+0.66 \cdot \text{height}}$. Increasing height by 1 inch *multiplies* the odds by $e^{0.66} = 1.94$; i.e., increases odds by 94%. Increasing height by 2 inches multiplies the odds by $e^{2 \cdot 0.66} = 3.76$; i.e., 276% increase.

The basic logistic regression model

Interpreting the model results

Evaluating the model

Logistic regression with multiple predictors

Interactions in logistic regression

Hypothesis testing

Applications



How good is our model?

- The usual R^2 isn't available for logistic regression.



How good is our model?

- The usual R^2 isn't available for logistic regression.
- There are “pseudo- R^2 ” metrics, but they're harder to interpret.



How good is our model?

- The usual R^2 isn't available for logistic regression.
- There are “pseudo- R^2 ” metrics, but they're harder to interpret.
- We'll focus on something simple: how many cases did we get right?



How many cases did we accurately predict?

Rule: predict male if $\widehat{P(\text{male})} \geq 0.5$, else female.

```
predicted.sex <- ifelse(predict(model, type="response") >= 0.5, 1, 0)
correct <- ifelse(predicted.sex == my.profiles$sex, 1, 0)
sum(correct) / nrow(my.profiles)

[1] 0.83
```




Comparing to a baseline

83% accuracy. Compare to always predicting the majority (male):

```
xtabs(~ sex, data=my.profiles) %>% prop.table()
```

```
sex  
  f   m  
0.4 0.6
```

Our model gives a “lift” from 60% (baseline) to 83%.



The confusion matrix

Pick one category as “positive” (here: male) and the other as “negative” (female). Every case falls into: TP, TN, FP, FN.



The confusion matrix

		Actual	
		Positive	Negative
Predicted	Positive	TP	FP
	Negative	FN	TN

TP = true male predicted male; TN = true female predicted female; FP = female predicted male; FN = male predicted female.



The confusion matrix

```
tab_conf <- xtabs(~ predicted.sex + my.profiles$sex)
tab_conf
```

	my.profiles\$sex	
predicted.sex	f	m
f	19466	5494
m	4623	30243

```
fn <- tab_conf[1,2]
```

```
fp <- tab_conf[2,1]
```

5494 false negatives (true M predicted F) and 4623 false positives (true F predicted M).



Rates from the confusion matrix

- True positive rate = $TP/(TP+FN)$: proportion of actual positives predicted positive
- True negative rate = $TN/(TN+FP)$: proportion of actual negatives predicted negative
- False positive rate = $FP/(TN+FP)$; False negative rate = $FN/(TP+FN)$



Confusion matrix (normalized)

```
xtabs(~ predicted.sex + my.profiles$sex) %>% prop.table(2)
```

```
      my.profiles$sex  
predicted.sex  f      m  
f 0.81 0.15  
m 0.19 0.85
```

Use `predicted ~ actual`; `prop.table(2)` normalizes by column (actual).



Model evaluation: final thoughts

- You can trade off FP vs FN by changing the threshold (predict 1 when $\hat{p} > p$ for $p \neq 0.5$).
- FP and FN can have very different costs (e.g., loans, medical tests).
- These are **in-sample** measures—optimistic. Use train/test or cross-validation for out-of-sample accuracy.

The basic logistic regression model

Interpreting the model results

Evaluating the model

Logistic regression with multiple predictors

Interactions in logistic regression

Hypothesis testing

Applications



Recap and adding predictors

Single-predictor model: $\log(\text{odds}) = -44.45 + 0.66 \cdot \text{height}$. We can add predictors just like in linear regression. Each coefficient is the effect on *log odds* (or odds) holding others constant.

```
model2 <- glm(male ~ height + orientation, data=my.profiles, family=binomial)
```



Adding another predictor

orientation has three categories:

```
xtabs(~ orientation, data=my.profiles)
```

```
orientation
```

bisexual	gay	straight
2763	5568	51495

Call:

```
glm(formula = male ~ height + orientation, family = binomial,  
     data = my.profiles)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-46.08076	0.37167	-124.0	<2e-16 ***
height	0.66535	0.00537	124.0	<2e-16 ***
orientationgay	2.09556	0.07209	29.1	<2e-16 ***
orientationstraight	1.39972	0.06068	23.1	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 80654 on 59825 degrees of freedom
Residual deviance: 43722 on 59822 degrees of freedom
AIC: 43730

Number of Fisher Scoring iterations: 6



Interpreting coefficients (multiple predictors)

$$\log \left(\frac{p}{1-p} \right) = -46.08 + 0.67 \cdot \text{height} + 2.1 \cdot \text{gay} + 1.4 \cdot \text{straight}.$$

- Intercept: predicted log odds for bisexual, height 0.



Interpreting coefficients (multiple predictors)

$$\log \left(\frac{p}{1-p} \right) = -46.08 + 0.67 \cdot \text{height} + 2.1 \cdot \text{gay} + 1.4 \cdot \text{straight}.$$

- Intercept: predicted log odds for bisexual, height 0.
- Among same orientation, each extra inch of height multiplies odds by $e^{0.67} = 1.95$ (95% increase).



Interpreting coefficients (dummies)

- Same height, gay vs bisexual: odds multiplied by $e^{2.1} = 8.13$ (713%).



Interpreting coefficients (dummies)

- Same height, gay vs bisexual: odds multiplied by $e^{2.1} = 8.13$ (713%).
- Same height, straight vs bisexual: odds multiplied by $e^{1.4} = 4.05$ (305%).



Understanding what's going on

```
xtabs(~ sex + orientation, data=my.profiles)
```

	orientation		
sex	bisexual	gay	straight
f	1994	1586	20509
m	769	3982	30986

```
xtabs(~ sex + orientation, data=my.profiles) %>% prop.table(2)
```

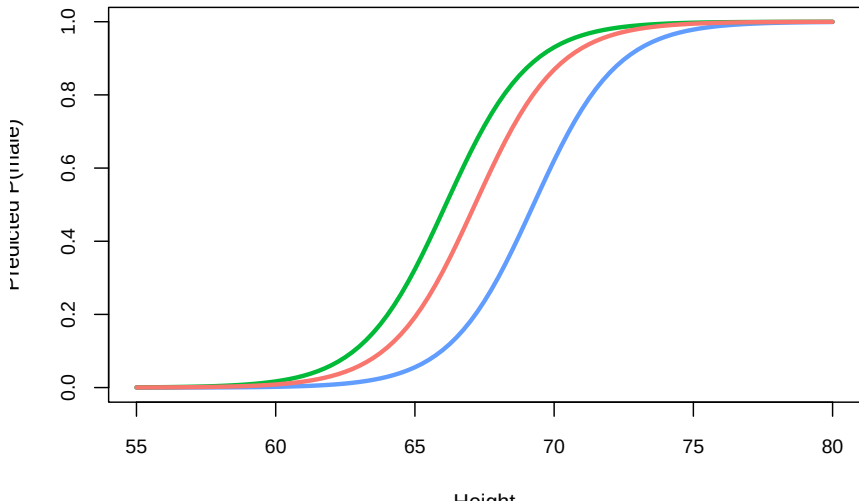
	orientation		
sex	bisexual	gay	straight
f	0.72	0.28	0.40
m	0.28	0.72	0.60





Three curves by orientation

Straight (red), gay (green), bisexual (blue):



The basic logistic regression model

Interpreting the model results

Evaluating the model

Logistic regression with multiple predictors

Interactions in logistic regression

Hypothesis testing

Applications



In linear regression, an interaction means the **slope** of X_1 depends on the **value** of X_2 . Same in logistic regression: the effect of X_1 on the *odds* depends on X_2 . Try height $\times$ orientation.

```
int.model <- glm(male ~ height * orientation, data=my.profiles, family=binomial)
summary(int.model)
```

Call:

```
glm(formula = male ~ height * orientation, family = binomial,
     data = my.profiles)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-35.3027	1.4050	-25.13	< 2e-16	***
height	0.5076	0.0206	24.67	< 2e-16	***
orientationgay	-6.2727	1.8365	-3.42	0.00064	***
orientationstraight	-10.2887	1.4596	-7.05	1.8e-12	***
height:orientationgay	0.1218	0.0271	4.49	7.1e-06	***
height:orientationstraight	0.1712	0.0214	8.01	1.2e-15	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 80654 on 59825 degrees of freedom
Residual deviance: 43663 on 59820 degrees of freedom
AIC: 43675

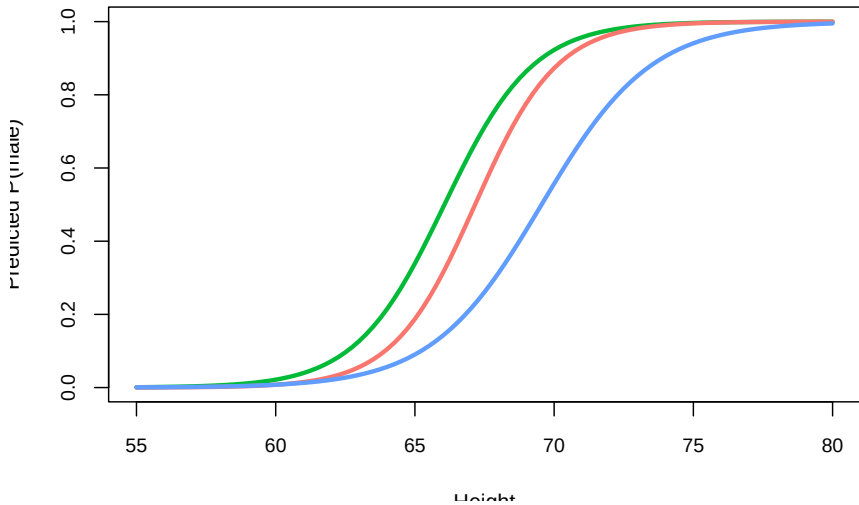
Number of Fisher Scoring iterations: 6



The interaction model

$$\log(p/(1-p)) = -35.3 + 0.51 \cdot \text{height} + -6.27 \cdot \text{gay} + -10.29 \cdot \text{straight} + 0.12 \cdot \text{height} \cdot \text{gay} + 0.17 \cdot \text{height} \cdot \text{straight}.$$

Gay (green), straight (red), bisexual (blue)—different slopes:



The basic logistic regression model

Interpreting the model results

Evaluating the model

Logistic regression with multiple predictors

Interactions in logistic regression

Hypothesis testing

Applications



Hypotheses to test

1. **Overall:** $\beta_1 = \beta_2 = \dots = 0$ (no predictive power)



Hypotheses to test

1. Overall: $\beta_1 = \beta_2 = \dots = 0$ (no predictive power)
2. One quantitative predictor: $\beta_i = 0$ (no effect of that X , holding others constant)

Hypotheses to test

1. **Overall:** $\beta_1 = \beta_2 = \dots = 0$ (no predictive power)
2. **One quantitative predictor:** $\beta_i = 0$ (no effect of that X , holding others constant)
3. **One dummy:** $\beta_i = 0$ (no difference vs reference level, holding others constant)



Overall null: likelihood ratio test

The likelihood ratio test compares the full model to an intercept-only model. Here: $\chi^2 = 36932$, $p < 0.001$ (highly significant).



Testing one coefficient

For a quantitative variable (e.g., height) or a dummy (e.g., straight vs bisexual), use the z and p -value for that coefficient in `summary(model2)`.

```
summary(model2)
```

```
orientationstraight    1.39972    0.06068    23.1    <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 80654  on 59825  degrees of freedom
Residual deviance: 43722  on 59822  degrees of freedom
```

The basic logistic regression model

Interpreting the model results

Evaluating the model

Logistic regression with multiple predictors

Interactions in logistic regression

Hypothesis testing

Applications



What else can we use logistic regression for?

- **Finance:** Predicting which customers are likely to default on a loan
- **Advertising:** Predicting response to a campaign
- **Marketing:** Predicting purchase or sign-up